Netflix Data Science Internship Project

Project Overview

This project analyzes the Netflix dataset using Python and data science techniques. It covers data cleaning, exploratory data analysis, content-based recommendations, and trend prediction.

Tasks Completed

Task 1: Data Cleaning and Preprocessing

- Loaded the Netflix dataset using Pandas.
- Identified missing values and duplicate records.
- Cleaned text columns and converted date values.
- Exported the cleaned dataset as a CSV file.

Task 2: Exploratory Data Analysis

- Analyzed Movies and TV Shows.
- Identified countries contributing the most content.
- Explored popular genres.
- Visualized release-year trends.

Task 3: Recommendation System

- Combined relevant Netflix metadata.
- Used TF-IDF for text feature extraction.
- Applied cosine similarity to identify similar titles.
- Generated content-based recommendations.

Task 4: Trend Prediction

- Analyzed Netflix content additions by year.
- Visualized historical trends.
- Trained a Linear Regression model.
- Generated preliminary estimates for future content additions.

Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Matplotlib

- Seaborn
- Scikit-learn

Conclusion

This project demonstrates the use of Python and machine learning techniques for data preprocessing, exploratory analysis, recommendation systems, and preliminary trend prediction.